Solutions

Modules 1–8

Problem 1 — Signals

Alpha rhythm: $f = 10\,Hz$, amplitude $30\,\mu V$. (a) Period in ms? (b) Cycles per second?

Answer. (a) Period $T = \dfrac{1}{f} = \dfrac{1}{10\text{ Hz}} = 0.1\text{ s} = \mathbf{100}$ ms.
(b) Frequency means cycles per second, so $f = 10$ cycles per second $\Rightarrow$ **10** full cycles in one second.

Why: Frequency and period are reciprocals. A 10 Hz wave does ten full cycles per second, so each cycle takes $1/10$ *of a second.*

Problem 2 — Sampling

Sampling rate $f_s = 256\,Hz$. (a) Highest faithfully recorded frequency? (b) A 200 Hz EMG burst aliases to what frequency?

Answer. (a) The Nyquist frequency is half the sampling rate:

$$f_{\text{Nyquist}} = \frac{f_s}{2} = \frac{256}{2} = \mathbf{128}\text{ Hz}.$$

Any signal at or below 128 Hz is faithfully recorded.
(b) 200 Hz is above Nyquist, so it folds back into the recorded band. The aliased frequency is:

$$f_{\text{alias}} = |f_s - f| = |256 - 200| = \mathbf{56}\text{ Hz}.$$

Why: Anything above the Nyquist frequency is misrepresented as a lower frequency that the digitizer can *represent. For a single tone f with $f_s/2 < f < f_s$, the alias appears at $f_s - f$.*

Problem 3 — Filtering

(a) Filter for 50 Hz mains noise? (b) Risk of an aggressive low-pass at 15 Hz?

Answer. (a) A **notch (band-stop) filter** centered at 50 Hz. It removes a narrow band around 50 Hz without disturbing nearby brain frequencies.
(b) You would miss **spikes** (and other fast activity). Spikes contain high-frequency components above 15 Hz, so a 15 Hz low-pass smooths them away. Clinically dangerous because spikes are a hallmark of **epilepsy** — missing them means missing the seizure focus.

Why: Filters can't tell brain signal from noise — they remove everything in the band you specify. The standard clinical low-pass is 35–70 Hz precisely so spike content is preserved.

Problem 4 — FFT

4 s recording at $f_s = 250$ Hz. (a) Samples? (b) Whole-segment Δf? (c) 1 s-window Δf?

Answer. (a) Number of samples = duration × sampling rate:

$$N = T \cdot f_s = 4 \text{ s} \cdot 250 \text{ Hz} = \mathbf{1000} \text{ samples.}$$

(b) FFT frequency resolution is the inverse of segment duration:

$$\Delta f = \frac{1}{T} = \frac{1}{4 \text{ s}} = \mathbf{0.25} \text{ Hz.}$$

(c) For 1-second windows:

$$\Delta f = \frac{1}{1 \text{ s}} = \mathbf{1} \text{ Hz.}$$

Why: Longer segments resolve finer frequency differences but blur in time. Shorter segments give crisp time localization but coarser frequency resolution. This is the time–frequency tradeoff (Module 7 expands on it).

Problem 5 — EEG Frequency Bands

Match brain state → rhythm → band.

Answer.

State	Rhythm	Band (Hz)
Deep sleep (N3)	Delta (δ)	1–4
Drowsy	Theta (θ)	4–8
Awake, eyes closed	Alpha (α)	8–13
Focused mental task	Beta (β)	13–30
Active thinking / cognition	Gamma (γ)	>30

Why: Slower rhythms reflect synchronous activity across large neuron populations (deep sleep). Faster rhythms come from local, asynchronous processing (gamma during active cognition).

Problem 6 — The 10-20 Electrode Naming System

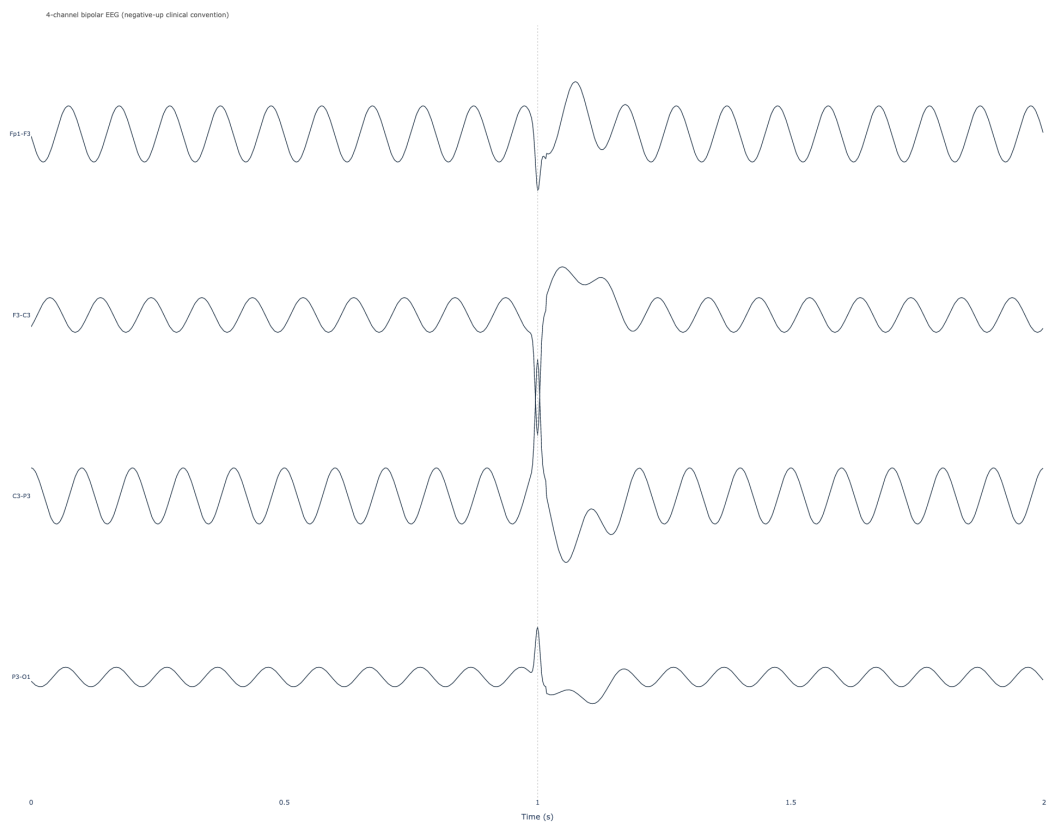
Identify the brain region and side for Fp2, T3, Pz, O1.

Answer.

Electrode	Brain region	Side
Fp2	Frontopolar (forehead)	**Right**
T3	Temporal	**Left**
Pz	Parietal	**Midline**
O1	Occipital	**Left**

Why: The letter encodes the lobe (Fp = frontopolar, F = frontal, T = temporal, C = central, P = parietal, O = occipital). The number encodes the side: ***odd numbers = left****,* ***even numbers = right****,* ***z = midline (zero)****.*

Problem 7 — Phase Reversal



(a) Electrode closest to the source? (b) How can you tell?

Answer. (a) **C3**.
(b) The two channels that share C3 — **F3-C3** and **C3-P3** — deflect in opposite directions at the spike (one downward, one upward). This is a **phase reversal**. The shared electrode of a phase reversal points to the source.

Why: In a bipolar montage each channel is the difference $V_A - V_B$. A focal source under one electrode produces opposite-sign contributions in any pair of channels that include it.

Problem 8 — The Double Banana

(a) List the four electrode chains of the longitudinal bipolar (double banana) montage. (b) What kind of brain activity does this montage tend to underestimate?

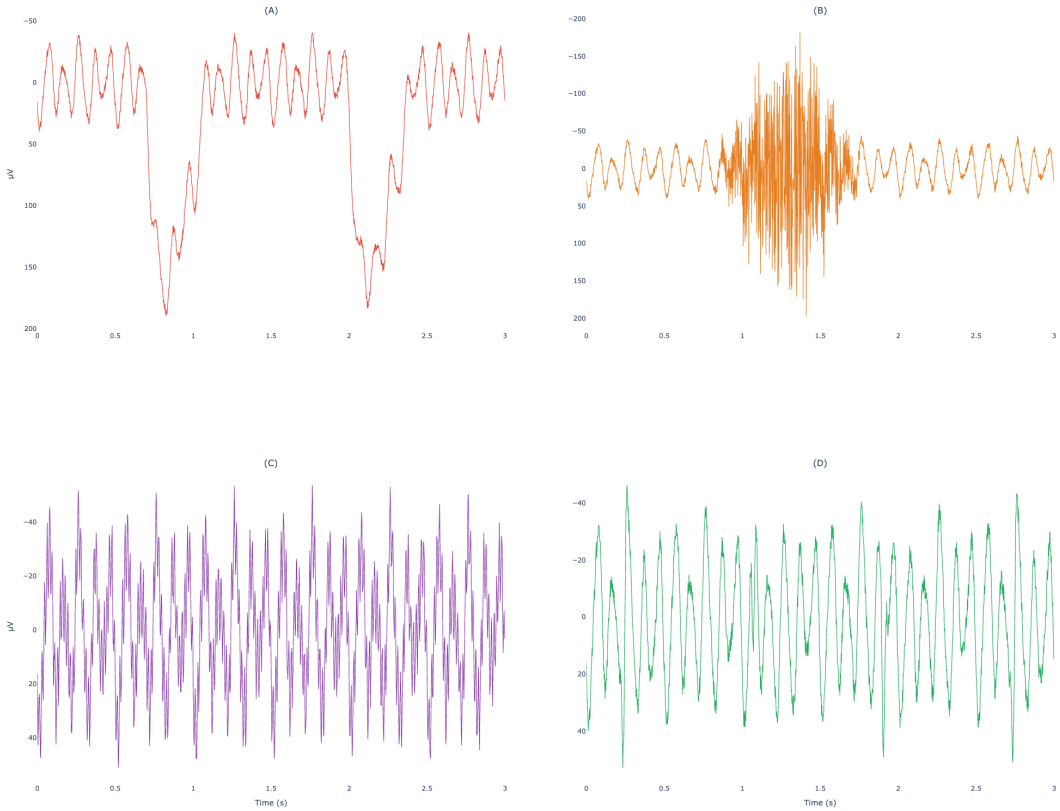
Answer. (a) Four front-to-back chains:

- **Left parasagittal:** Fp1 → F3 → C3 → P3 → O1
- **Right parasagittal:** Fp2 → F4 → C4 → P4 → O2
- **Left temporal:** Fp1 → F7 → T3 → T5 → O1
- **Right temporal:** Fp2 → F8 → T4 → T6 → O2

(b) **Lateral / inter-hemispheric activity** — differences between left and right at the same anterior-posterior level. Because all four chains run front-to-back, side-to-side voltage differences never appear as a phase reversal; they smear across the chain instead. The **transverse bipolar montage** (left-to-right chains) is used to catch this.

Why: Each chain only sees voltage gradients along its own direction. A purely lateral asymmetry has no front-to-back gradient, so the double banana under-represents it. Different montages reveal different patterns — which is why neurologists switch montages while reading.

Problem 9 — Artifacts



Identify each panel; pick one and give a removal strategy.

Answer.

Panel	Artifact
A	**Eye blink** — large slow asymmetric deflection
B	**Muscle (EMG)** — high-frequency broadband fuzz, burst-like
C	**60 Hz line noise** — perfectly regular, dense sinusoid
D	**ECG (cardiac)** — sharp periodic spikes at ∼1.2 Hz

Removal (any one is sufficient):

- *Blink:* Independent Component Analysis (ICA) to subtract the blink component, or ask the patient to fixate on a point.
- *Muscle:* Ask the patient to relax the jaw and face; low-pass filter (caveat: loses real high-frequency brain activity).
- *60 Hz:* Notch filter centered at 60 Hz; check electrode impedances.
- *ECG:* Record a simultaneous ECG channel and subtract its projection.

Why: Each artifact has a distinctive morphology and spectral signature. Recognizing them is the first step — only then can you remove them or reject the contaminated epoch.

Problem 10 — Spectrograms

Window 0.5 s vs 2 s: (a) what improves vs worsens? (b) which for seizure-onset detection?

Answer. (a) Going 0.5 s → 2 s:

- Frequency resolution **improves** (sharper). $\Delta f = 1/T$, so it goes from $1/0.5 = 2$ Hz down to $1/2 = 0.5$ Hz.
- Time resolution **worsens**. Each time slice now covers 2 s instead of 0.5 s, so events are smeared horizontally.

(b) Use the **shorter (0.5 s) window**. Seizure onset must be localized to within a fraction of a second; you accept coarser frequency precision in exchange for sharp temporal precision.

Why: Time–frequency uncertainty: $\Delta f = 1/T$. You cannot simultaneously have fine frequency resolution and fine time resolution. Sharp temporal events demand short windows; slow-rhythm characterization (sleep staging) tolerates long windows.

Problem 11 — The Real Thing (Module 8)

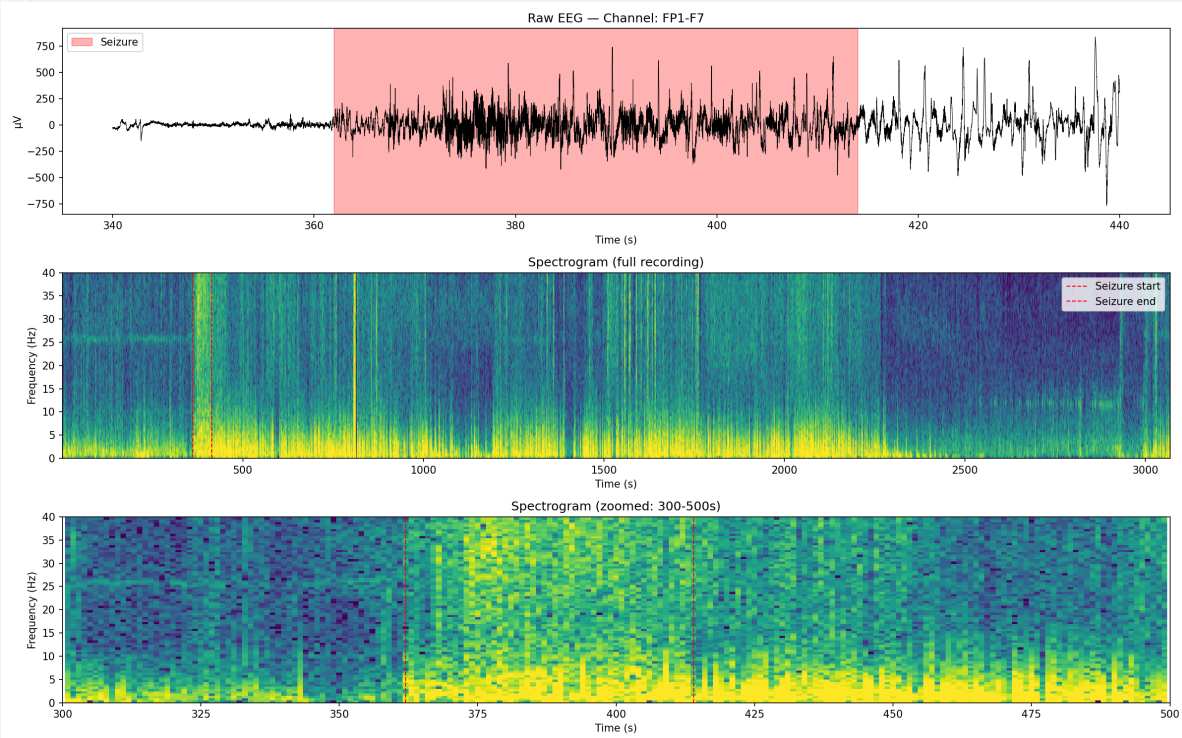
Load `chb03_01.edf` *from the CHB-MIT Scalp EEG Database. Answer five questions.*

Answer. (a) **23 channels.** Sampling rate: **256 Hz.** Duration: **∼51.1 minutes** (3068 seconds).
Note: The EDF header says 3600 s but the actual recording is shorter.

(b) **Bipolar montage.** The channel names are electrode pairs (e.g., FP1-F7, F7-T7, T7-P7), not individual electrodes. Each channel is the voltage difference between two adjacent electrodes.

(c) Seizure onset at approximately **362 seconds**, ending at approximately **414 seconds**. Duration: **∼52 seconds**.

(d) See verification plot below.



The spectrogram shows a sudden broadband power increase starting at ∼362 s, spanning 0–40 Hz, lasting about 52 seconds. This is consistent with electrographic seizure activity.

(e) The patient is having an **electrographic seizure**. The EEG shows sudden onset of rhythmic activity with broadband power increase at 362 seconds, lasting 52 seconds. Combined with the clinical history of staring spells with lip-smacking, this is consistent with **focal seizures** (likely temporal lobe epilepsy).

Why: The CHB-MIT dataset annotates this seizure at 362–414 s. The spectrogram makes it obvious: a clear transition from normal background to broadband rhythmic activity, then gradual post-ictal slowing. The bipolar montage is identifiable from channel names alone — no metadata lookup required.